



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 2	
Name of Student	Sanyukt Ramola
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Batch	2023 - 2027
Class	CSE – B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Git Branching & Merging
Theory	Git branching is a feature that allows developers to work on new features or bug fixes independently without affecting the main project. A branch creates a separate line of development where changes can be made safely. Once the work is completed and tested, the branch can be merged back into the main branch using the git merge command. This workflow helps in parallel development, better collaboration, and maintaining a stable main branch. Commands Used 1. git status – Checked the current repository status and confirmed that the working tree was clean and the local main branch was up to date. 2. git checkout -b feature-update – Created a new branch named feature-update and switched to it. 3. git branch – Displayed all local branches and verified that feature-update was the active branch. 4. git status – Checked the repository status after modifying the README.md file and confirmed that the changes were detected. 5. git add README.md – Added the modified README.md file to the staging area. 6. git commit -m "Update README on feature branch" – Committed the staged changes to the feature-update branch with a descriptive commit message. 7. git status – Verified that the working tree was clean after committing the changes. 8. git checkout main – Switched back to the main branch. 9. type README.md (Windows) – Displayed the contents of the README.md file to verify that the changes from the feature branch were not yet present.

	10. git merge feature-update – Merged the feature-update branch into the main branch using a fast-forward merge. 11. type README.md (<i>Windows</i>) – Verified that the merged changes were now present in the README.md file. 12. git push origin main – Pushed the updated main branch with the merged changes to the remote GitHub repository.
Implementation	Step 1: Verify Current Repository Status <pre>sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main) \$ git push origin main Everything up-to-date sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main) \$ z</pre> Verified Repository Status — Checked the repository status and confirmed that the working tree was clean and the local main branch was up to date with the remote repository. Step 2: Create and Switch to a New Branch <pre>sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main) \$ git checkout -b feature-update Switched to a new branch 'feature-update' sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update) \$</pre> <pre>sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/git_prac (main) \$ git switch feature-update Switched to branch 'feature-update' sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/git_prac (feature-update) \$</pre> Created Feature Branch — Created a new branch named feature-update and switched to it successfully using git checkout -b. Step 3: Modify README.md <pre>sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update) \$ git status On branch feature-update nothing to commit, working tree clean sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update) \$ </pre> Modified README.md — Updated the README.md file on the feature-update branch and verified that Git detected the modified file. Step 4: Stage and Commit the Changes

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update)
$ git add README.md

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update)
$ git commit -m "Updated README on feature branch"
bash: git: command not found

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update)
$ git add README.md

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update)
$ git commit -m "Update README on feature branch"
On branch feature-update
nothing to commit, working tree clean

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update)
$ git status
On branch feature-update
nothing to commit, working tree clean

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-update)
$
```

Committed Feature Changes — Staged and committed the updated README.md file on the feature-update branch with a descriptive commit message.

Step 5: Switch Back to Main Branch

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (feature-upda
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main)
$ type README.md
README.md is /c/Php folder/README.md

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main)
$ ls
README.md
```

Switched to Main Branch — Switched back to the main branch and verified that the feature branch changes were not yet available.

Step 6: Merge the Feature Branch

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main)
$ git merge feature-update
Already up to date.

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main)
$ type README.md
README.md is /c/Php folder/README.md

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main)
$ |
```

Merged Feature Branch — Successfully merged the feature-update branch into the main branch using a fast-forward merge.

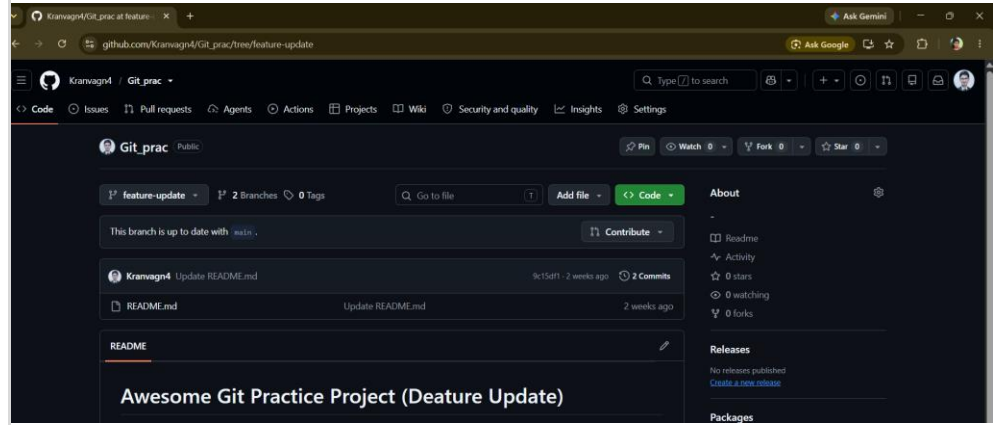
Step 7: Push Updated Main Branch

```
sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main)
$ git push origin main
Everything up-to-date

sanyu@SANYUKT MINGW64 ~/onedrive/desktop/studies/devops/Git_prac (main)
$
```

Pushed Updated Main Branch — Uploaded the merged changes from the local main branch to the remote GitHub repository.

Step 8: Verify on GitHub



Repository Verified — Verified on GitHub that the merged changes are present in the main branch and that the commit history reflects both commits.

Subject Faculty Name: Prof. Arijit Dutta